Normal distribution :

Notation: $X \sim N(\mu, \sigma)$

where, $\mu = E(X)$ & $\sigma = \text{std}(X)$.

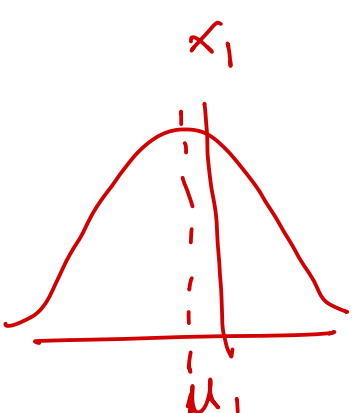
pdf : $f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$

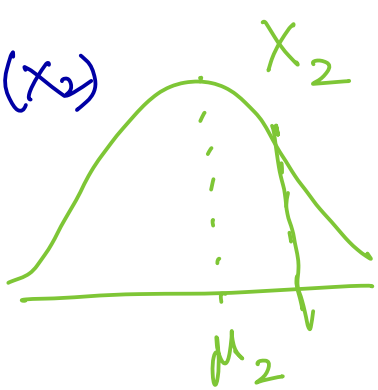
$N(\mu_1, \sigma_1)$ $N(\mu_2, \sigma_2)$

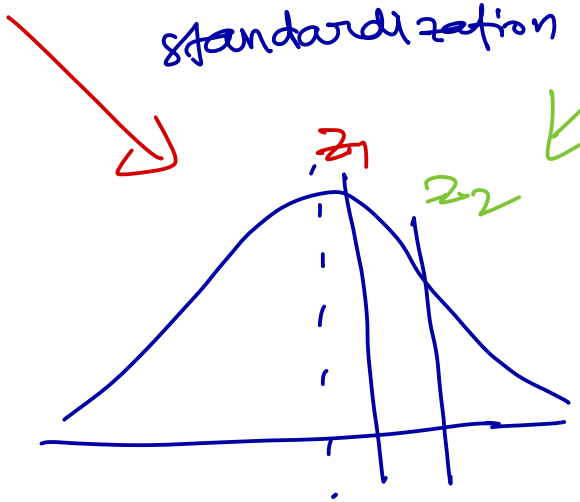
X_2

X_1

$\text{pnorm}(x_1) < \text{pnorm}(x_2)$







$$z_1 = \frac{x_1 - \mu_1}{\sigma_1}$$

$$z_2 = \frac{x_2 - \mu_2}{\sigma_2}$$

z-scores → standardized value of a normal random variable

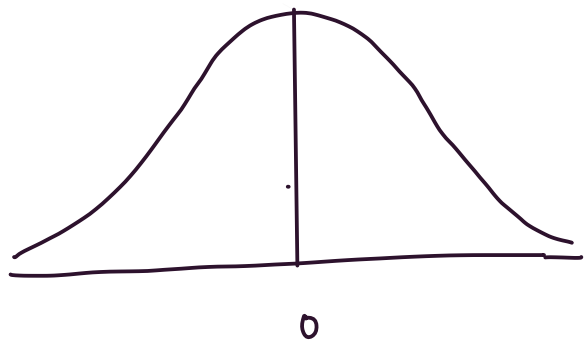
$$z = \frac{x - \mu}{\sigma} \sim N(0,1)$$

$N(0,1)$ → standard normal distribution


y = f(x) = density of x.
b
a
f(x) dx
a
1

Standard Normal distribution

$N(0,1)$.



Standardization:

$$N(\mu, \sigma) \longrightarrow N(0,1).$$

x → value of the r.v.

μ → $E(x)$ value

σ → $std(x)$ value.

$$\frac{x-\mu}{\sigma}$$

X: The height of male students in STAT 240

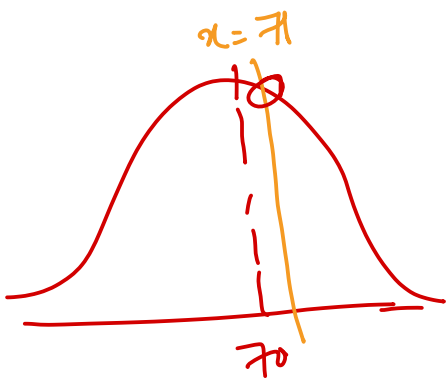
Y = height of female students in STAT 240.

$X \sim N(\mu_x, \sigma_x)$

$\mu_x = 70$ in

$\sigma_x = 3$ in

$x = 71$ in



P dnorm (71, 70, 3)

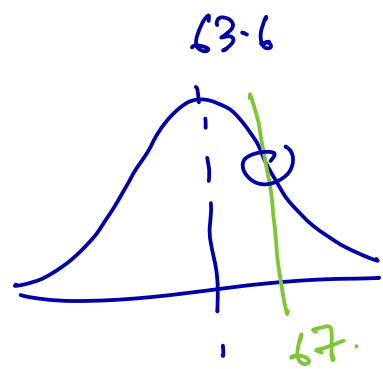
$= 0.63$

$Y \sim N(\mu_Y, \sigma_Y)$

$\mu_Y = 63.6$ in

$\sigma_Y = 2.5$ in.

$y = 67$ in



P dnorm (67, 63.6, 2.5).

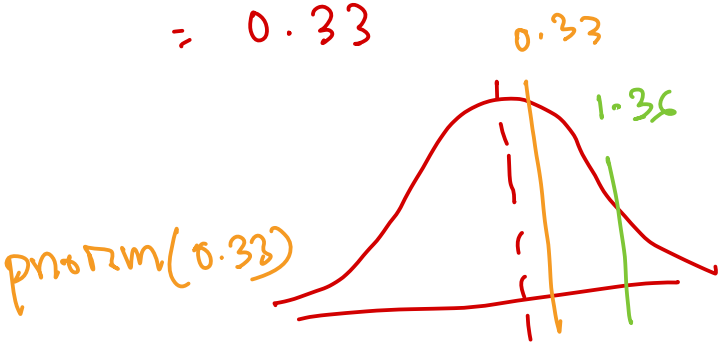
$= 0.91$

$$z_x = \frac{71-70}{3}$$

$$= 0.33$$

$$z_y = \frac{67-63.1}{2.5}$$

$$= 1.36$$



$= pnorm(71, 70, 3)$

$pnorm(1.36)$

$= pnorm(67, 63.1, 2.5)$

Central Limit Theorem (CLT)

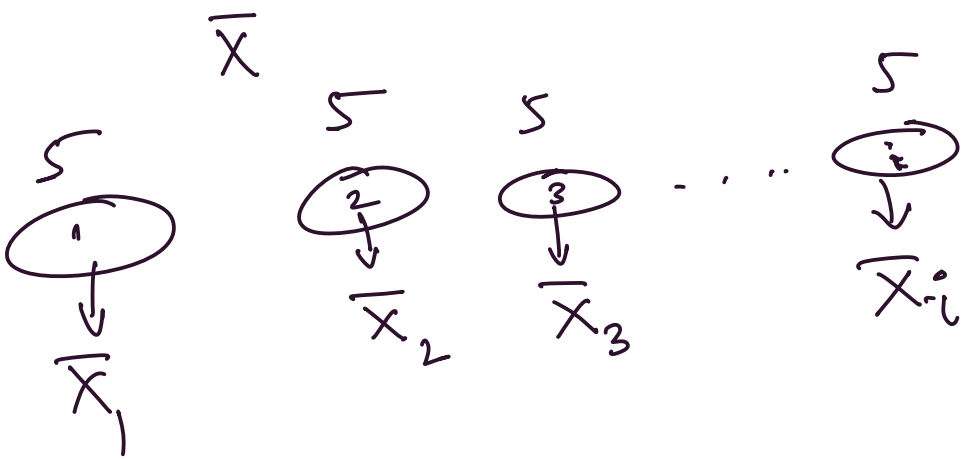
$$X \sim D$$

$$E(X) = \mu$$

$$std(X) = \sigma$$

take a sample of size n of X.

$$\bar{X} = \frac{1}{n} \sum x_i$$



$$\bar{X} \sim N\left(\mu, \frac{\sigma}{\sqrt{n}}\right)$$

$X \sim D(\mu, \sigma)$ then $\bar{X} \sim N\left(\mu, \frac{\sigma}{\sqrt{n}}\right)$

↑ random variable

↑ sample mean of a sample of size n.

$\frac{\sigma}{\sqrt{n}}$ → sampling error.

if X is a non-normal dist

$n \geq 30$.

if X is a normal r.v

then $\bar{X}$ is always $N\left(\mu, \frac{\sigma}{\sqrt{n}}\right)$